

## SUB-STRAND 3.2: CURRENT ELECTRICITY

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*Grade 10 Physics - Electricity and Magnetism Strand*

### Sub-Strand Overview

| Component                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| Specific Learning Outcomes        | By the end of this sub-strand (22 lessons), learners will:<br>a) Explain terminologies used in current electricity<br>b) Verify relationships: $V=IR$ , $E=I(R+r)$<br>c) Determine resistance and resistivity of various conductors using different methods<br>d) Determine effective resistance of various resistor networks<br>e) Determine the relationship of potential difference and current to power (heating effect)<br>f) Appreciate applications of current electricity in day-to-day life |
| Key Inquiry Questions             | 1. How is current electricity applicable in our day-to-day life?<br>2. Why are resistors used in electrical circuits?                                                                                                                                                                                                                                                                                                                                                                                |
| Core Competencies                 | <ul style="list-style-type: none"><li>• Critical Thinking and Problem Solving</li><li>• Learning to Learn</li><li>• Communication and Collaboration</li><li>• Digital Literacy</li></ul>                                                                                                                                                                                                                                                                                                             |
| Values                            | <ul style="list-style-type: none"><li>• Responsibility</li><li>• Respect</li><li>• Integrity</li><li>• Unity</li></ul>                                                                                                                                                                                                                                                                                                                                                                               |
| Pertinent and Contemporary Issues | <ul style="list-style-type: none"><li>• Safety and Security: Electrical safety in laboratories and homes</li><li>• Life Skills: Problem-solving through experiments</li><li>• Education for Sustainable Development: Energy efficiency</li></ul>                                                                                                                                                                                                                                                     |

# LESSON 1: INTRODUCTION TO CURRENT ELECTRICITY

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Sub-Strand 3.2 - Lesson 1 of 22

## A. Specific Learning Outcomes

| Category                          | Outcomes                                                                                                                                                                                                                                                                                                                                                                                                             |
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| Knowledge                         | <ul style="list-style-type: none"><li>• Understand that electric current is the flow of electric charge through a conductor</li><li>• Know that electrical energy can be transformed into other forms of energy (heat, light, motion)</li><li>• Recognize that different electrical devices produce different amounts of heat</li><li>• Understand basic electrical safety principles in Kenyan households</li></ul> |
| Skills                            | <ul style="list-style-type: none"><li>• Observing and comparing electrical phenomena</li><li>• Predicting outcomes based on prior knowledge</li><li>• Analyzing patterns in energy transformations</li><li>• Making connections between science concepts and daily life</li></ul>                                                                                                                                    |
| Attitudes                         | <ul style="list-style-type: none"><li>• Appreciation for electrical safety</li><li>• Curiosity about how everyday devices work</li><li>• Wonder at energy transformations in technology</li></ul>                                                                                                                                                                                                                    |
| Science and Engineering Practices | <ul style="list-style-type: none"><li>• Asking questions about electrical phenomena</li><li>• Developing models of current flow and heating</li><li>• Planning investigations systematically</li><li>• Constructing evidence-based explanations</li></ul>                                                                                                                                                            |

## B. Overview

| Element              | Details                                                                                                                                                                                                                     |
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| Key Inquiry Question | Why do some electrical devices get hot when plugged in? Where does this heat energy come from?                                                                                                                              |
| Purpose in Storyline | This lesson launches the electricity unit by presenting a puzzling everyday phenomenon that students have experienced. Students observe that phone chargers, kettles, and other devices become warm or hot during use. This |

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|                     | drives inquiry into electric current, energy transformation, and resistance.                                                                                                                                                                                                                                                               |
| <b>Safety Notes</b> | <ul style="list-style-type: none"> <li>• CRITICAL: Never touch exposed electrical wires or open plugs</li> <li>• Only touch plastic casings of devices, never metal parts</li> <li>• Unplug devices before examination</li> <li>• Report any damaged electrical equipment immediately</li> <li>• Discuss KPLC safety guidelines</li> </ul> |

### C. Lesson Implementation Framework

| Learner Experience                                                                                                                                                                                            | Resource Link                                                                                                                                                                                | Teacher Moves                                                                                                                                                                                                                                                                                                   | Sensemaking Strategy                                                                                                                                                                                                                   | Formative Assessment                                                                                                                                                                                                                                                                                                |
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| <b>PREDICT Phase:</b><br>Teacher shows phone charger plugged in 30+ minutes. Students feel it (safely). They predict WHY it's warm and WHERE heat comes from. Draw initial models of what's happening inside. | <ul style="list-style-type: none"> <li>• Phone charger</li> <li>• Prediction sheets</li> <li>• Drawing materials</li> <li>• DQB chart paper</li> </ul> <p>[ARES VIDEO: Electric Current]</p> | <p>Facilitating Prediction:</p> <ul style="list-style-type: none"> <li>• "What do you notice?"</li> <li>• PROBING: "Why warm?"</li> <li>• WAIT TIME: 10 seconds</li> <li>• "Draw what's happening"</li> </ul> <p>Establishing DQB:</p> <ul style="list-style-type: none"> <li>• Record ALL questions</li> </ul> | <p>POE - PREDICT:</p> <p>Think-Pair-Share:</p> <ul style="list-style-type: none"> <li>• Think (2 min)</li> <li>• Pair (3 min)</li> <li>• Share with class</li> </ul> <p>DQB Started</p>                                                | <ul style="list-style-type: none"> <li><input type="checkbox"/> Makes prediction</li> <li><input type="checkbox"/> Provides reasoning</li> <li><input type="checkbox"/> Draws model</li> <li><input type="checkbox"/> Contributes to DQB</li> </ul> <p>Misconception Alert:<br/>Note incorrect ideas</p>            |
| <b>OBSERVE Phase:</b><br>Students observe different devices (kettle, fan, LED lamp, charger, iron). Record observations: Device name, Gets hot?, What does it do?, How hot?                                   | <ul style="list-style-type: none"> <li>• Various devices</li> <li>• Infrared thermometer</li> <li>• Data sheets</li> </ul> <p>[ARES SIMULATION: Device Energy]</p>                           | <p>Facilitating Observation:</p> <ul style="list-style-type: none"> <li>• "Which hottest? Coolest?"</li> <li>• PROBING: "What patterns?"</li> <li>• WAIT TIME: Full observation</li> <li>• Circulating among groups</li> </ul>                                                                                  | <p>POE - OBSERVE:</p> <p>Create data table:<br/>Device   Hot?   Purpose</p> <p>Think-Pair-Share:</p> <ul style="list-style-type: none"> <li>• Patterns?</li> <li>• Predictions correct?</li> </ul>                                     | <ul style="list-style-type: none"> <li><input type="checkbox"/> Accurate data</li> <li><input type="checkbox"/> Identifies patterns</li> <li><input type="checkbox"/> Compares to predictions</li> </ul> <p>Quick Check:<br/>"Which two hottest?"</p>                                                               |
| <b>EXPLAIN Phase:</b><br>Students develop explanations for WHY some devices get hot. Discussion: What IS electricity? Where does energy come from? Where does it go?                                          | <ul style="list-style-type: none"> <li>• Explanation scaffolds</li> <li>• Discussion prompts</li> <li>• Energy diagrams</li> </ul> <p>[ARES VIDEO: Energy Transform]</p>                     | <p>Facilitating Explanation:</p> <ul style="list-style-type: none"> <li>• "WHY do some get hot?"</li> <li>• PROBING: "Where does heat come from?"</li> <li>• WAIT TIME: 15 seconds</li> <li>• "Build explanation together"</li> </ul>                                                                           | <p>POE - EXPLAIN:</p> <p>Think-Pair-Share:</p> <ul style="list-style-type: none"> <li>• Think (3 min)</li> <li>• Pair (4 min)</li> <li>• Share</li> </ul> <p>Build understanding:<br/>Electricity = moving charges carrying energy</p> | <ul style="list-style-type: none"> <li><input type="checkbox"/> Uses evidence</li> <li><input type="checkbox"/> Scientific reasoning</li> <li><input type="checkbox"/> Mentions transformation</li> <li><input type="checkbox"/> Connects to phenomenon</li> </ul> <p>Exit Ticket:<br/>"Why does charger warm?"</p> |

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| <b>Kenya Context:</b><br><b>Discussion: How does understanding electricity help Kenya? KPLC's role, cost (KSh/kWh), energy efficiency, solar panels on rural schools, electrical safety, careers.</b>                | <ul style="list-style-type: none"> <li>• KPLC posters</li> <li>• Electricity bills</li> <li>• Solar panel photos</li> </ul> <p>[ARES VIDEO: Electrical Safety Kenya]</p> | <b>Facilitating Connection:</b> <ul style="list-style-type: none"> <li>• "How used in Kenya?"</li> <li>• PROBING: "Why understand safety?"</li> <li>• WAIT TIME: 8 seconds</li> <li>• "What careers?"</li> </ul>                    | <b>Think-Pair-Share:</b> <ul style="list-style-type: none"> <li>• KPLC provides power</li> <li>• Solar in rural areas</li> <li>• Costs money (KSh)</li> <li>• Safety important</li> <li>• Careers available</li> </ul> | <input type="checkbox"/> Identifies KPLC role<br><input type="checkbox"/> 2+ Kenya applications<br><input type="checkbox"/> Shows safety awareness<br><input type="checkbox"/> Career connections<br><br>Quick Check:<br>"Why safety important?" |
| <b>Model Revision:</b><br><b>Students return to initial models and revise based on new evidence. Show: device, current flowing, energy transformation, where heat comes from. Write explanation of changes made.</b> | <ul style="list-style-type: none"> <li>• Initial models</li> <li>• Revision materials</li> <li>• Colored pencils</li> </ul> <p>Comparison: Before/After</p>              | <b>Facilitating Revision:</b> <ul style="list-style-type: none"> <li>• "What evidence now have?"</li> <li>• PROBING: "How show energy changing?"</li> <li>• WAIT TIME: 10 seconds</li> <li>• "Scientists revise models!"</li> </ul> | <b>Model Revision:</b><br><br>Show: <ol style="list-style-type: none"> <li>1. Electrical source</li> <li>2. Current flow</li> <li>3. Device</li> <li>4. Energy transform</li> <li>5. Clear labels</li> </ol>           | <input type="checkbox"/> Shows transformation<br><input type="checkbox"/> Labels components<br><input type="checkbox"/> Explains changes<br><input type="checkbox"/> Uses evidence<br><br>Note: "Model Revised"                                  |

#### D. Teacher Reflection

1. Did POE cycle engage students? Which phase strongest?
2. Did I ask probing questions with sufficient wait time (10-15 sec)?
3. Are DQB questions posted visibly?
4. What do initial models reveal about student understanding?
5. Did students connect to Kenyan contexts (KPLC, costs, careers)?
6. Do all students understand electrical safety?
7. Student Feedback: "What surprised you most?" (Record 5-10 responses)

#### E. Summary Table Prompt

Teacher says: "Add Lesson 1 to your Summary Table."

Prompt for students:

- What did I observe? (Devices, patterns. Note: DQB Started, Model Created, Model Revised)
- What did I learn? (Key concepts about electricity and energy)
- How does this explain the phenomenon? (Why does charger get warm?)